Data Science Toolbox Portfolio Questions

09 Parallelism

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Block 9

Portfolio 09

Portfolio 09.1.1 - Vectorisation in pytorch

- Why has modern machine learning converged on tensor operations (matrix products, convolutions, element-wise functions)?
  - Link vectorisation to automatic differentiation, and how GPUs can compute this "quickly".

Portfolio 09.2.1 - Computational Graphs

- How are complex computations structured in machine learning? What are the basic operations that make optimisation automatable?
  - Reason about the role of manual structuring of computation. When is it still useful for a data scientist to worry about it?

Portfolio 09.3.1 - Big Compute vs Big Data

- You are working on a problem for which there is a (potentially expensive) **simulator** and a (potentially inexpensive) **emulator**, which can estimate what the simulator will output, and can be updated from the simulator. You can run several simulator parameters (say 64) in parallel.
  - What choices impact how you might use the emulator to choose good parameter choices?
  - An example is given for a Random Forest. The Forest is our simulator that we are trying to model, and the regression is an emulator for its performance.